METHODOLOGY
MSCI SUSTAINABILITY AND CLIMATE

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MSCI ESG Ratings Methodology: Opportunities in Clean Tech Key Issue

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Introduction

This document provides essential information on the components of the Opportunities in Clean Tech Key Issue to enable users of ESG Ratings to understand how MSCI Sustainability & Climate's (MSCI S&C's)[1] outputs are determined. Opportunities in Clean Tech is a Key Issue in the Environmental Pillar of the MSCI ESG Ratings model. Companies are evaluated on their positioning to meet market demand for products with environmental benefits.

For additional details on the MSCI ESG Ratings Model, refer to the Data, ratings and scores section of "ESG Ratings Methodology."

Exhibit 1: MSCI ESG Key Issue hierarchy

3 Pillars	10 Themes	33 ESG Key Issues
Environmental	Climate Change	Carbon Emissions
		Climate Change Vulnerability
		Financing Environmental Impact
		Product Carbon Footprint
	Natural Capital	Biodiversity & Land Use
		Raw Material Sourcing
		Water Stress
	Pollution & Waste	Electronic Waste
		Packaging Material & Waste
		Toxic Emissions & Waste
	Environmental Opportunities	**Opportunities in Clean Tech**
		Opportunities in Green Building
		Opportunities in Renewable Energy
Social	Human Capital	Health & Safety
		Human Capital Development
		Labor Management
		Supply Chain Labor Standards
	Product Liability	Chemical Safety
		Consumer Financial Protection
		Privacy & Data Security
		Product Safety & Quality
		Responsible Investment
	Stakeholder Opposition	Community Relations
		Controversial Sourcing
	Social Opportunities	Access to Finance
		Access to Health Care
		Opportunities in Nutrition & Health

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3 Pillars	10 Themes	33 ESG Key Issues
Governance	Corporate Governance	Board
		Pay
		Ownership & Control
		Accounting
	Corporate Behavior	Business Ethics
		Tax Transparency

Opportunities associated with this Key Issue

- Increased access to markets due to regulatory reforms and incentives.
- Increased revenue from capturing changing demand, early-mover advantage.

Opportunities in Clean Tech Key Issue score

The Opportunities in Clean Tech Key Issue score evaluates the company's level of exposure to, and management of, opportunities on this Key Issue. The Key Issue score is based on the Exposure score and the Management score using the formula below. The Exposure score and the Management score are combined such that the Exposure score determines the weight placed on the Management score, ranging from 50% (when Exposure score is equal to zero) to 100% (when Exposure score is equal to 10).

$$KI_i = \left(\left(\frac{EXP_i}{20} + .5\right) \times MGMT_i\ Score\right) + \left(\left(.5 - \frac{EXP_i}{20}\right) \times 5\right)$$

Where:

- KI_i is the Key Issue score for company i.
- EXP_i is the Exposure score for company i.
- $MGMT_i$ is the Management score for company i.

Opportunities in Clean Tech Management framework

The table below outlines the indicators mapped to their respective components that inform the Opportunities in Clean Tech Management score. MSCI S&C assigns each indicator a default weight. Indicator weights may be adjusted at the company level to reflect the relevance of the indicator to the company's business activities. If a company has no exposure to the business activities covered by an indicator, the indicator is assigned a weight of zero.

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Component	Sustainability Indicator
Opportunities in Clean Tech Strategy score	Targets to increase investment in clean tech
	Strategic focus on clean technology development
Opportunities in Clean Tech Programs & Initiatives score	Research and development intensity (as percentage of sales) score
Opportunities in Clean Tech Performance score	Clean tech revenue score

Opportunities in Clean Tech Management score

The Opportunities in Clean Tech Management score evaluates the company's ability to manage its exposure to opportunities on this Key Issue. It is based on weighted average of the score associated with each relevant indicator.

This score is derived from indicators that are scored on a 0-10 scale, with 10 corresponding to best practice. Refer to the Analyzing risk management section of "ESG Ratings Methodology" for additional details. Sources are company disclosures except where otherwise indicated.

The following formula is used to calculate the Opportunities in Clean Tech Management score:

$$MGMT_i = \sum Indicator_Score_{j,i} * Indicator_Weight_{j,i}$$

Where:

- $MGMT_i$ is the Management score - Excluding Controversies for company i.
- $Indicator_Score_{j,i}$ is the score of indicator j for company i.
- $Indicator_Weight_{j,i}$ is the weight of indicator j for company i.

The following formula is used to calculate the scores for the components

$$COMP_{c,i} = \sum_{j \in c} Indicator_Score_{j,i} * Indicator_Weight_Scaled_{j,i}$$

Where:

- $COMP_{c,i}$ is the component score for component c, for company i.
- $Indicator_Score_{j,i}$ is the score of indicator j for company i.
- $Indicator_Weight_Scaled_{j,i}$ is the scaled weight of indicator j for company i.

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- $j \in c$ indicates that the summation is over all indicators belonging to component c.

The following formula is used to calculate the weights for the components

$$COMP_Weight_{c,i} = \sum_{j \in c} Indicator_Weight_{j,i}$$

Where:

- $COMP_Weight_{c,i}$ is the component weight for component c for company i.
- $Indicator_Weight_{j,i}$ is the weight of indicator j for company i.
- $j \in c$ indicates that the summation is over all indicators belonging to component c.

The definitions of the data points in each component are given below:

Opportunities in Clean Tech Strategy score

The Opportunities in Clean Tech Strategy score evaluates the company's initiatives to embed clean technology into its long-term strategy through business focus, R&D direction, and investment targets. It is expressed as a 0–10 score, where 10 indicates strong initiatives.

- **Targets to increase investment in clean tech**

Definition: Indicates whether the company has disclosed quantitative targets to increase investment in clean technology.

- **Strategic focus on clean technology development**

Definition: Assesses the company's strategic commitment to clean technology as part of its core business focus and growth plans. Highest performance reflects clean technology being a core business line or a primary focus for future R&D and innovation. Expressed as a 0–10 score, where 10 indicates the highest level of performance.

Opportunities in Clean Tech Programs & Initiatives score

The Opportunities in Clean Tech Programs & Initiatives score evaluates the company's initiatives to advance clean technology through research and development. It is expressed as a 0–10 score, where 10 indicates strong initiatives.

- **Research and development intensity (as percentage of sales) score**

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Definition: Assesses the company's performance on R&D intensity metrics (including R&D intensity normalized by sales, and trend in R&D intensity) relative to its peers. Expressed as a 0–10 score, where 10 indicates the highest level of performance.

- **R&D intensity(% of sales) industry relative score**

  Definition: Assesses the company's performance on R&D intensity (as a percentage of sales) relative to its industry peers. Expressed as a 0-10 score, where 10 indicates strong performance.

  - **Three-year average R&D intensity(% of sales)**

    Definition: Measures the company's three-year average R&D intensity (as a percentage of sales).

- **R&D intensity (% of sales) industry relative trend score**

  Definition: Assesses the company's three-year average of year-over-year R&D intensity (as a percentage of sales) trend, relative to its peers. Expressed as a 0–10 score, where 10 indicates strong performance.

  - **Three-year average R&D intensity trend (% of sales)**

    Definition: Measures the company's three-year average of year-over-year R&D intensity **(**as a percentage of sales) trend.

Opportunities in Clean Tech Performance score

The Opportunities in Clean Tech Performance score evaluates the company's performance on clean technology metrics (including revenue generation from clean technology products and services). It is expressed as a 0–10 score, where 10 indicates strong performance.

- **Clean tech revenue score**

Definition: The percentage of the company's revenue from clean tech activities. Expressed as a 0-10 score, where 10 indicates that the company derives more than 50% of its revenue from clean technology products and services.

Opportunities in Clean Tech Exposure score

The Opportunities in Clean Tech Exposure score evaluates the company's exposure to opportunities on this Key Issue. It is based on the Business Exposure score only. The Business Exposure score is scored on a 0-10 scale, with 10 corresponding to the highest opportunity and 0 corresponding to the lowest

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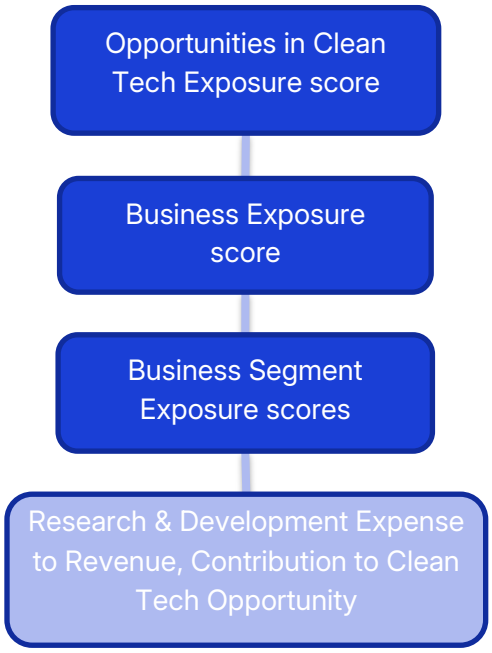
opportunity. Refer to the Analyzing risk exposure section of "ESG Ratings Methodology" for additional details. The following formula is used in the Exposure score calculation:

$$EXP_{OCT,i} = BUS_{OCT,i}$$

Where:

- $EXP_{OCT,i}$ is the Opportunities in Clean Tech Exposure score of company i.
- $BUS_{OCT,i}$ is the Business Exposure score of company i.

Exhibit 2: Exposure score components



Business Exposure score

The Business Exposure score is a weighted average of the Business Segment Exposure scores of a company's business segments. scores are weighted by the proportion of a company's total revenue in each business segment, using the following formula to calculate the Business Exposure score:

$$BUS_{OCT,i} = \sum_{j=1}^{n_i} w_{Revenue,i,j} BSE_{OCT,j}$$

Where:

- $BSE_{OCT,j}$ is the Business Segment Exposure score for business segment j.
- $w_{Revenue,i,j}$ is the weight of business segment j for company i based on the the contribution to total company revenue.

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- n_i is the number of business segments of company i.

Business Segment Exposure score

The Business Segment Exposure score of a business segment is based on a mapping of the business segment as disclosed by the company to a corresponding business activity. MSCI S&C uses the Standard Industrial Classification (SIC) system along with industry-specific adjustments to define business activities. The score associated with a business activity is used to calculate a Business Segment Exposure score. Each business activity score is determined by a weighted average of the Research and Development Expense score of the activity and Contribution of Opportunities in Clean Tech score of the activity:

$$BSE_{OCT,j} = 0.3RDF_j + 0.7COCT_j$$

Where:

- RDF_j is the Research and Development Expense score for business activity j.
- $COCT_j$ is the Contribution of Opportunities in Clean Tech score for business activity j.

A business segment's Research and Development Expense score is assessed using the ratio of research and development expenses to revenues at the business activity level. The Research and Development Expense score is scored on a 0-10 scale, with 10 corresponding to the highest research and development expenses per unit of revenue and 0 corresponding to the lowest research and development expenses per unit of revenue.

The Contribution of Opportunities in Clean Tech score of a business activity is assessed using a qualitative assessment based on predefined guidance to determine whether the business activity is a major contributor to opportunities in clean technology. Segments which are major contributors receive a score of 10. Segments with no potential opportunities in clean technology receive a score of 0.

Data sources

The following sources are used to determine the Opportunities in Clean Tech Exposure score:

Sources: Refinitiv, MSCI S&C, company disclosures.

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